Servers for BT's interactive TV services

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Servers are a major component in an interactive TV service, and once systems of only modest size are considered, the technology requirements are challenging. If BT moves to a roll-out of broadband interactive TV services, then close working with chosen suppliers will be required if costs are to be viable.

This paper looks at some server architectures and technologies that might eventually be part of BT's interactive TV service.

1. Introduction

The task of storing, managing, and spooling out information for BT interactive TV services is not trivial, once reasonable numbers of customers are using the services — in the Market Trial (1995) system, 2000 hours of audiovisual material requires nearly 2 Tbyte of on-line storage, with a potential of 1250 independently controlled 2-Mbit/s streams (a total of 2.5 Gbit/s) from the one server.

The majority of BT's interactive TV services will be based on stored information, on a server or set of servers, and this paper looks at various ways of meeting the requirements for various types of services that might be eventually bracketed under the umbrella of 'BT Interactive TV'.

2. What is a server?

A typical server (see Fig 1) consists of four major logical parts:

- audiovisual database, where the audiovisual files, and possibly their associated embedded control and graphics information, are stored,
- audiovisual information processor and pump, where the information to be spooled to customers is extracted from the audiovisual database, formatted where required, and sent to the network interface for transmission to the user — in addition, this entity also would probably handle the bit-level loading of information from a live feed, or a mass storage device, under the control of the server,

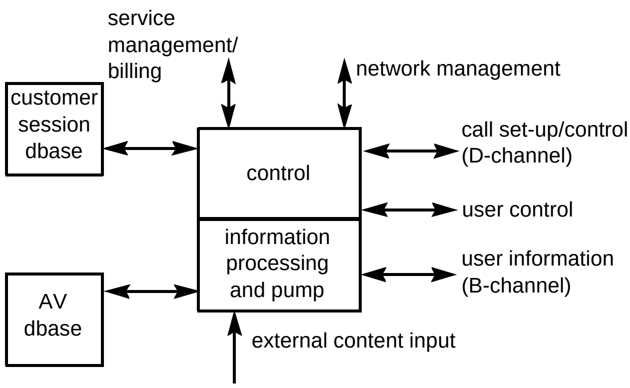

Fig 1 A typical server.

- server control, which manages the server, its sessions, and its databases, and provides the intelligence to handle each user session, including session requests and clear down, while at the same time providing the intelligence to handle requests from the network management systems and service management systems, and packaging information for passing to the billing system,
- the customer session database, which is used by the server control to handle each user on an individual basis according to the various parameters dictated by the overall service and its status, this specific programme, and the individual users' preferences.

A server can be considered to have the following interfaces:

- stored information sent to user (and in some situations received from users),
- control channel to/from users, so they can control their sessions,
- call set-up/control to/from network,
- network management, so status and integrity of server can be remotely managed by the server provider,
- service management/billing, to permit the service provider to enable and disable services and customers' access to them, extract billing and use this information, and control the overall service provided,
- external content input, where new material can be loaded on to the server under the control of the service provider — this material might additionally or exclusively be available in real time as a broadcast feed, where permitted; the material will, in general, include audio-visual information, graphics and navigation/control information.

3. Server parameters

Various main requirements on a server will tend to indicate different technical solutions. The major requirements (see Fig 2) are considered to be:

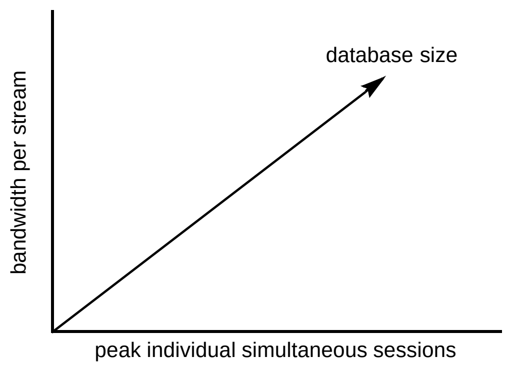

Fig 2 The major requirements on a server.

In addition, the level of interactivity for each session will also have a bearing on the solutions provided.

3.1 Bandwidth per stream

Low bit rate (LBR)

LBR is taken to mean each stream is of bandwidth $n \times$ 64 Kbit/s, where n = 1—6. Such servers operating at these speeds already exist for speech applications in the BT network, and can also be readily used for facsimile databases and telemetry systems. With few changes, such servers could also be used for database services for ISDN videophone and ISDN multimedia terminals, such as the PC-based VC8000. The LBR servers can interface directly to the BT transmission network either as a peripheral (for example connected to a series of standard ISDN lines) or as part of the network (30-channel multiplex and CCITT No. 7 signalling).

Such servers could be used as part of a BT interactive TV service where:

- LBR audiovisual quality is acceptable through the use of still images, graphics or carefully chosen material content, or
- the server acts as a controller of the customer session and the audio-visual signal is transmitted via another system.

Broadband (HBR)

Broadband is taken here to mean where each stream is $n \times$ 1 Mbit/s, with n = 2—8. This bit rate can provide entertainment quality TV (or better), but at present widely deployed networks do not exist which can handle such streams to users. Other papers in this edition address the issues of broadband connections to users.

The BT Interactive TV Market Trial (1995), and the earlier product development trial (1994) both use broadband streams at a total bit rate of 2.048 Mbit/s per stream. This bit rate is considered to give sufficiently good quality for most film and TV material which has been encoded off-line, but some customers may wish to pay more for EDTV or even HDTV quality in the future.

3.2 Peak traffic (sessions)

As far as the server is concerned, this parameter relates to the peak simultaneous independent sessions that it is required to support. BT is currently only permitted to carry fully one-to-one entertainment services over its network, so, in that situation, one independent stream corresponds to one user. In systems used mostly for transactional services, this is not a real restriction, but, for systems focusing on the film/TV programmes services, some bunching of users might be technically possible in order to reduce costs — this bunching is variously known as narrowcasting, or simulcasting, and the benefit to the provider is an increase in users per server stream.

For a given performance, the parameter of peak simultaneous sessions has a major impact on the server costs, as it affects:

- network interface, as the larger the number of peak simultaneous sessions at a given bit rate, the higher the capacity of the network interface required,
- bit-carrying power within the audio-visual information processor and pump,
- organisation of storage, as a higher traffic requirement will mean a greater aggregate sustained data rate from the overall audio-visual storage system,
- performance of the server control sub-system, as more processing power will be required to provide a given quality of service to users as the number of peak sessions increases.

It is therefore not surprising that early servers from suppliers have tended to provide a maximum of 100-200 simultaneous sessions at ~ 2 Mbit/s per stream, because, beyond that, special design of some of the various server sub-systems is required.

3.3 Audio-visual database size

For a given bit rate per stream, the database size (in gigabytes) clearly is related directly to the amount of material to be stored (in terms of hours of material). Conversely, the reverse is also obvious — for a given number of hours of material, the size of the database (in gigabytes) is directly related to the bit rate of the streams to users. However, the amount of raw data storage required is also dependent on implementation details, such as:

- replication of information in order to meet the requirement for the peak number of sessions to be supported,
- use of derived material from the raw source in order to reduce real time processing needs, e.g. visible fast-forward or fast-reverse files,
- use of redundant information in order to protect the on-line data from most likely storage media failures,
- storage fragmentation, as a result of material being updated or churned,
- buffer storage in order to carry out storage defragmentation,
- use of space to store new material prior to the old material being deleted and the new material being made available to users.

Various technologies can provide cost-effective solutions for the different service scenarios, and these will be discussed later in the paper. While considering the design of the database system, the trends in cost/performance of raw data storage for various storage media need to be considered carefully.

3.4 Interactivity

Little and Venkatesh [1] categorise interactive services into the following five groupings, based on the amount of interactivity allowed:

- broadcast (no video on demand (VoD)) services similar to broadcast TV, in which the user is a passive participant and has no control over the session,
- pay-per-view (PPV) services in which the user signs up and pays for specific programming, similar to existing CATV PPV services,
- quasi-video on demand (Q-VoD) services, in which users are grouped based on a threshold of interest — users can perform rudimentary temporal control activities by switching to a different group,
- near-video on demand (N-VoD) services, in which functions like forward and reverse are simulated by transitions in discrete time intervals (of the order of 5 minutes) this capability can be provided by multiple channels with the same programming skewed in time,
- true video on demand (T-VoD) services, in which the user has complete control over the session presentation, having full-function VCR (virtual VCR) capabilities, including forward and reverse play, freeze, and random positioning — T-VoD needs only a single channel (multiple channels become redundant) and is sometimes called interactive VoD (I-VoD).

T-VoD services can probably also be subdivided into:

- films on demand — users choose the item of interest and then control their viewing as if on their local VCR,
- browsing and transactional services, such as home shopping.

Near VoD is also sometimes called ‘staggercast’.

In the one extreme case of broadcast, the interactivity level is zero, so the demands on the server are negligible; at the other extreme of transactional services, the demands on the server will be similar to that on a public database service, and the processing requirements for such servers, when dealing with many thousands of simultaneous sessions, are considerable. The demands of interactivity will mostly effect the server control, and so will tend to be independent of bit stream bandwidth to the user.

3.5 Other server parameters — server sizing and location

An overall service to users does not necessarily have to be provided by a single server platform, and in many situations it may be more cost effective to use a multiplicity of platforms, especially where the users are scattered over a large geographical area. In addition, for a company such as BT, there is the option to consider the servers as being part of the telecommunications network, or simply as attached peripherals to the network. These parameters are considered here.

Network-based servers

Network-based servers can either be provided on the basis of a few large centralised servers for a large geographical area (for example, the UK), or many small and localised servers serving smaller numbers of users each. Both approaches have their advantages, and will be discussed below.

When considering network-based servers, in order to reduce operational costs it is vital that they have high availability, and can be remotely managed to the network management requirements of the service provider. In addition, they will probably need to provide a standard interface to the service provider's service management and billing systems. Servers in the network can be configured to be 'intelligent peripherals' as part of the intelligent network, or a formal attachment to the network and yet reside on the service provider's premises. Low bit rate servers can use existing signalling systems (such as CCITT No. 7) and so fit into the existing network structure; broadband servers will have to meet emerging signalling standards, such as the possibility of the ITU-T Recommendation Q.2931 as a peripheral.

In large systems, it has been suggested that servers be considered in a hierarchy, depending on the demands of users on their material; a more local server can pass on the request to a remote server if the information the user is requesting is not available. It needs to be noted that this approach can make considerable demands on the local server to handle the information from the remote site, and current modelling would indicate that this may not be cost-effective. However, different systems and services will demand different solutions. Irrespective of the number of servers provided, each one will need to be kept in date and in track with other servers, especially where material is replicated across servers. With a small number of large servers, the information tracking can be a fairly straight forward affair, but this does mean that high-bandwidth links are required from local 'points of presence' all the way through to these servers, with their associated costs, whereas when using smaller servers at local points of presence (for example the local exchange) the bandwidth of the main network to some form of 'update' server is going to depend entirely on the rate of update of information required, leading to some potential savings in transmission. The effects of having servers in local exchange buildings must not be underestimated, as the operational costs of providing maintenance cover across so many locations UK-wide will inevitably be large in order to provide an acceptable quality of service to end customers. Figures 3 and 4 show the two main options, although others obviously exist.

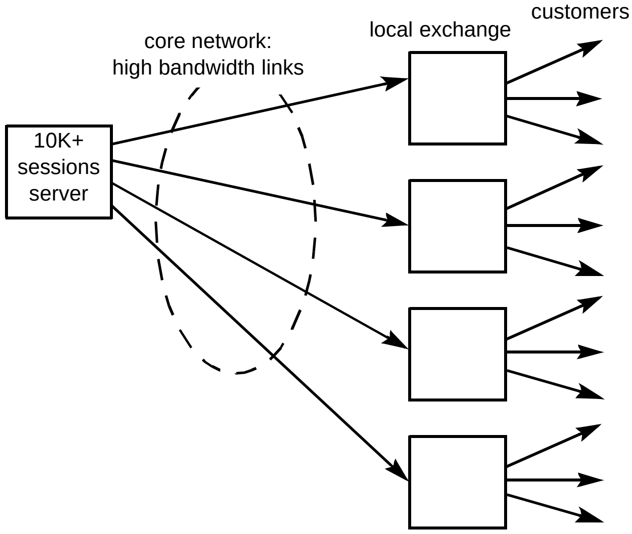

Fig 3 Large server within network.

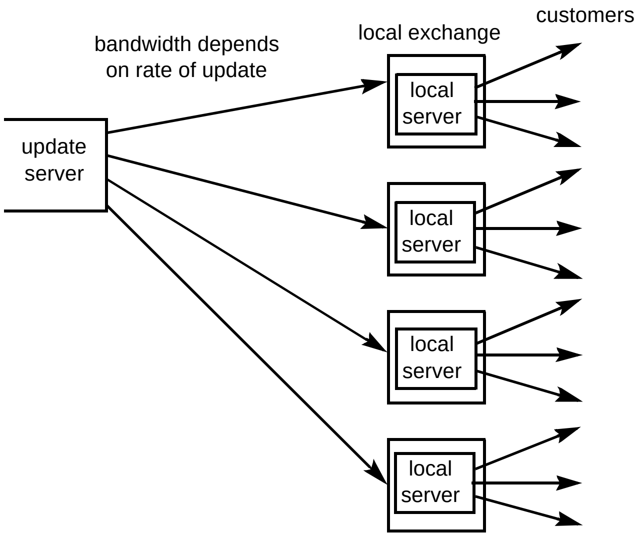

Fig 4 Local servers updated from remote system.

It must also be noted that when opting for the use of smaller local servers, there may well be greater up-front costs at the early stages of service deployment, as a completely functioning server is required at each local exchange where the first user is connected, whereas with a remote larger server, that server can be built up as the overall traffic increases. The main operational costs as far as the server is concerned will focus on support for the storage media, as these tend to involve mechanical items, and do not offer high reliability compared with all-electronic systems.

Customer-based servers

These servers tend to be smaller units, and are only ‘attachments’ to a network. They only require interfaces to meet the service requirements of the service provider, and will probably not be remotely managed, and so will not be part of any intelligent network or network management system. However, various solutions are already emerging for such smaller systems for the hotel, call centre, and airline entertainment markets.

4. Broadband servers — some of the family

Broadband servers tend to break down into various categories, depending on the level of interactivity required and the scale of the server, for example:

- large-scale, interactive for transactional multimedia services (10K-plus peak sessions),
- large-scale for TV programmes, films, interactive VoD only,
- large-scale for near-VoD type services,
- trial cable-TV and airline/hotel types (~ 200 sessions).

4.1 Interactive broadband server

A typical interactive broadband server for a large network system (see Fig 5) will probably address something like:

- 10—50K-plus simultaneous sessions,
- greater than 20 Gbit/s downstream capacity total,
- greater than 100 Mbit/s user control peak capacity,
- 2000-plus hours of material — greater than 2 Tbyte audio-visual store.

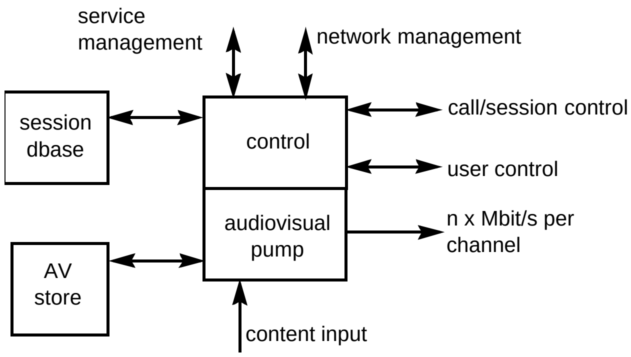

Fig 5 A typical interactive broadband server.

In contrast, a customer-based system will probably demand:

- 200 simultaneous sessions,
- greater than 400 Mbit/s downstream capacity,
- greater than 2 Mbit/s user control peak capacity,
- 200 hours of material — greater than 200 Gbyte audio-visual store.

The server control will depend on the level of interactivity of the service required — if the bulk of customers will be using transactional services, then the demands on the control will be far greater than those just used for films and TV programmes on demand.

Audio-visual store implementation

For an interactive broadband server, some material will be very popular at a particular time, other material will be rarely accessed, but, in most systems, users will demand that material is available ‘instantly’. This would tend to imply that slow media, such as tape, are not feasible for this service, especially if tapes are loaded mechanically on to a tape reader from a demand of a user (response times liable to approach ½ to 2 minutes). However, some implementors consider one solution to be to store the first few minutes of each title on a rapid-access storage medium (like magnetic disk) to cover the time taken to load the tape; others state that the cost/bandwidth of tape is not much different from disk once all aspects are considered, and focus entirely on disk-based systems. Some implementors believe that solid state (RAM) storage for popular items can reduce overall costs. Table 1 contains data from Little and Ventkatesh [1] regarding storage media.

Table 1 Comparative storage media costs [1].

Storage Type	Cost/Mbyte	Sessions/ Device	Cost/Movies stored
RAM	$50.00	200	$50,000
Hard disk	$0.50	5	$400
R/W optical	$0.20	2	$200
Magnetic tape	$0.01	1	$10

From the data in Table 1, comparative sessional costs can be derived (see Table 2).

Table 2 Comparative sessional storage media costs.

Storage Type	Cost/Session/Mbyte	Cost/Session/Film
RAM	$0.25	$250
Hard Disk	$0.1	$100
R/W Optical	$0.1	$100
Magnetic Tape	$0.01	$10

However, the tape and optical disk options exclude (presumably) the cost of the readers and any mechanical handlers, and so perhaps are not directly comparable with the others.

The majority of solutions to date for this type of broadband server have focused on the clever use of magnetic disk stores, either as one complete entity, or spread across multiple processor or spooling elements. In any sizeable disk store, the design needs to take into account maintaining the data integrity in the event of a failure of an individual disk drive, as the MTBF of disks is sufficiently finite to make failure of one drive in a large system quite a probable event over a period of months — this is not surprising, given the mechanical nature of disk drives. Techniques such as RAID [2] can be used which enable a faulty disk drive to be replaced and the data automatically rebuilt on to it, although clearly this is only possible where the original data has been stored with sufficient additional redundancy to enable the lost data to be derived without reloading from back-up.

In some scenarios, it is very hard to predict the demand patterns across the various parts of the database (for example, the popularity of a specific film or presentation), and the database then has to be so designed to meet these unknowns. One such technique is 'disk striping', where a given audio-visual item is scattered across a large number of disk drives, so a given user is constantly and invisibly moved from drive to drive while watching one item; each drive is filled with segments from many different audio-visual items, so the probability of demand on a drive can be fairly well defined. Clearly such a system demands very careful data management, and also the necessary algorithms to handle demand appropriately as it arrives.

In addition, in a number of services, material will need to be updated, old material deleted and new material loaded. This 'churn' process demands careful management of the database, in terms of:

- efficient use of database capacity, while providing continuous service to users,
- careful management of data bandwidths so that storing new material does not saturate the capabilities of the database system to provide service to users.

Turning to the processor elements that control and spool the data from the disks stored to the network interface, these elements need to be able to handle data very rapidly, but usually without performing additional processing. Suppliers to date are tending to focus on either producing bespoke hardware to perform this function, modifying some form of 'massively parallel processing' (MPP) system, or have a multiplicity of standard, smaller-scale processor systems with their own associated disks, and switching their stream outputs.

Proponents of MPP systems claim the following advantages:

- the CPU elements are mass produced, and therefore low cost — in some cases the MPP vendors use standard CPUs, so do not pay directly for the development of new generations of CPU, but use them as they become available,
- MPP architectures can inherently offer large overall bandwidths, as any one path does not carry large amounts of traffic,
- spare processor power can be used if required,
- expansion in terms of database size and/or simultaneous users is easily performed without major structural impact,
- no additional switching is required on the network side, as all switching is performed within the MPP system, so any stream can appear on any physical network interface.

Proponents of the bespoke solution would claim the following advantages:

- systems specifically designed to meet server requirements,
- no wasted system or processor power,

- more readily reduced in size, power consumption and cost as the market size increases and technology permits.

Proponents of the multiple small-scale processor systems tend to opt for PC (or equivalent) systems, claiming advantages as:

- PCs are commodity items and very low cost,
- expansion is relatively easy — you buy more PCs,
- very little hardware development cost.

The main drawback of some proposed systems lies in the need (or otherwise) of the server to supply any session on any system-defined network interface, not necessarily the interface that would be optimum for the server. To that end, some designs include a captive switch, especially where ATM-based network interfaces are used.

Server control

In many ways the control part of the server has opposite requirements compared with the audio-visual control — as the server controller is required to handle all the interactions with the customer, and control the overall server, it needs processing power and intelligence, with less emphasis on simple data handling.

Many vendors to date are therefore focusing on providing high-performance UNIX ™-based systems of a size dependent on processing power required. These systems are either standard UNIX workstations, or can be symmetric multiprocessing (SMP) or even MPP systems.

Clearly in some scenarios, an MPP vendor might choose to integrate the audio-visual pump processor/spooler with a server control, allocating processors with an overall MPP for the server control, some for the network interface, and some for the spooling activity; other vendors might choose to allocate processors on a more dynamic basis as instantaneous demand dictates. This, though, will require some highly sophisticated low-level software.

The main focus of the server control must be to handle large numbers of user (and associated call control) interactions with a good overall response time. Much of the user's perception of response time will depend on the overall software architecture used and hence the processing performance of the set-top box, but some of this processing is inevitably performed by the server, however sophisticated the set-top box. Again, careful sizing of the server control to meet the needs of peak simultaneous sessions, expected user interactions, and the software environment chosen, will be required if the user is to perceive a good quality of service, especially when using transactional services. Much of the sizing can be based on traditional sizing models for multi-user real time systems, and is not discussed further here.

So far, the server model used has assumed a single, logical element for the server control; in some situations there may be the potential of having a set of loosely coupled processors controlling one audio-visual database system, each processor handling a set of user sessions. This may be of particular importance if the user applications are highly processor-intensive for the server.

Content loading

At first glance, the loading of encoded audio-visual information and associated data on to the server is a trivial file transfer problem, but once consideration is given to the large amounts of data involved, it can be seen as in fact a non-trivial operation. Content can arrive in a number of ways:

- mass storage device, e.g. computer tape, containing a complete encoded audio-visual file to load,
- a communications link to a remote system, over which encoded audio-visual files are sent,
- a communications link from which a live or broadcast programme is received.

Common computer tape systems can spool at perhaps 6 Mbit/s, so for each hour of material encoded at 2 Mbit/s, approximately 20 minutes of loading time is required. An additional set-up time (to put the tape in the device, issue commands at the operator console, permit the tape to be checked by the processor before spooling, and at the end remove the tape) might be 5 minutes or more; so, assuming film/TV programmes of average duration 90 minutes, one tape drive might support the loading on average of 1.7 programmes/hour. If a server has 2000 hours of film/TV material, then at the start there might be 1333 programmes to load, which would take perhaps 780 hours to load on one tape drive. This corresponds to nearly 100 standard working days if one drive is used, with the server fully occupied performing this file loading.

In the situation of obtaining material over a communications link, much the same discussion applies as that from a tape, or other mass storage device, except that more checking is possibly required to ensure the information arriving is that which is expected, and is dealt with appropriately. The bandwidth of the communications link required will directly relate to the rate of material update required; higher-level protocols to use when sending programme information to a server are only just emerging from industry associations such as DAVIC (Digital Audio-Visual Council [3]. The server must also be sized to be able to accept and process the incoming information at the rate received in addition to continuing live service to users. Such processing may not be trivial if the local server is required, for example, to derive visible fast-forward and fast-reverse files from MPEG play files.

In the situation of content arriving from a live or broadcast programme feed, different criteria will apply. Broadcast feeds will clearly be one-way, and there will be no way the server can request flow control on the information, so it must be capable of receiving and dealing with the information as it arrives. In general, the information will be simply an encoded audio-visual stream, with no inherent content, graphics, user application, or menu data; these elements will have to be provided either separately or manually. As the broadcast feed may be received in unencoded form, a real time encoder and multiplexer will be required with appropriate hardware interface to the server. Currently, such encoders are relatively expensive. The server may be additionally required to make the broadcast feed immediately available to users, either on a ‘live’ basis, or on a basis of their being able to watch from the start-up to the current point of the programme, with use of some VCR controls as appropriate. Clearly in the BT situation, the current regulatory environment prohibits an entertainment broadcast feed being made available to users, so currently only the option to store the audio-visual information is permitted. Some form of additional control information will be required from the broadcast feed, so the server is aware of the actual start/end of programme segments or items, if they are to be stored for further use.

However, the loading of the material is not simply the audio-visual content, but must also include the categorisation of the material, menu routeing to that material, and text and graphical information on that material, so the eventual users can be directed to their choice. The actual mechanisms to be used for this are the responsibility of the applications software, but, while standards exist for audio-visual encoded files, they do not yet exist for all the ancillary information for films, and a lot of manual intervention is required. Usually this heavily increases the time taken to load fully a new programme on to the system.

While the problem of loading is perhaps most acute at service start-up, it does not disappear if there is to be a significant amount of programme churn on the server (i.e. new programmes being introduced and old ones being removed). As such churn will take place on the live server, it is vital that the process ensures that the integrity of service is maintained at all times to users, and that menus accurately reflect the content and programmes actually available to a user.

When transactional services are considered, additional information is required to be loaded, relating to the application-specific processing required by the service, whether that processing takes place in the server, set-top box, or on both. Much of the control information will be produced from an authoring package, by the information provider, but the audio-visual segments will probably need to be encoded separately and brought together with the other information to produce the complete service. A transactional service will therefore probably consist of many files, and the management of these while loading will be vital to the integrity of the service.

Content definition

Standards for the description and definition of files for server contents are currently being discussed in forums like DAVIC [3], but no decisions have been made on this at the time of writing. In general, contents being loaded on to a server will need to include definitions of:

- linear audio-visual segment (embedded audio with moving video, such as an MPEG multiplex) — parameters will be required to define encoding and multiplexing used,
- still-video image — parameters required for defining screen resolution and encoding system used (e.g. JPEG) and screen location/relative size,
- audio segment — parameters required to define encoding and any multiplexing used, and possibly, number of audio channels with the multiplex provided, and whether those channels are alternatives (e.g. multi-lingual) or multi-channel sound, or some mixture of the two,
- graphics object — parameters required to define encoding (and any multiplexing) used, intended relative/absolute screen position and overall size in relation to the overall screen size, whether overlaid or replacing any video and/or existing graphics on screen, whether any movement (time, vector) or change in shape (3-D vector and/or mesh definition) while displayed parameters are required to define the performance of additional software requirements (e.g. in the terminal),
- cursor/highlighting object — parameters to define when the cursor is used, its appearance (any graphics will need the definition of encoding supplied), any blinking rate, etc,
- execution instructions — parameters are required to define the condition(s) for the execution to occur and the commands which execute them,
- signal-state table — the complete set of actions required for all possible interactions (user, system, time) with the application at the application’s current state (will point to executables and/or display objects),

- execution map — a definition of the start and end states for the application and the top-end management programme for the given application,

- mass storage requirements — parameters will include the (perhaps contiguous) space required for the various objects within the content and the intended bit rate, if not implied from the encoded material object, so that sufficient mass storage bandwidth can be allocated within the server,

- execution requirements when running application — parameters will include a definition of the assumed requirements from the authoring tool, so that the receiving server can ensure that it loads appropriate run-time environments for the given executable (either or both on server and the sets of set-top boxes potentially connected to it), and, at the time of execution, can ensure that adequate server resources are allocated to the application — some default sets of execution parameters would be expected to be given which will work to the emerging industry-standard authoring tools at current versions, and for these to be updated in an agreed manner as tools are further developed,

- minimum requirements of user interface input device, for example, remote control with DAVIC-defined minimal set of keys, mouse, QWERTY keyboard, speech-driven system, 486 PC, Mac, etc,

- external communications/facilities required to run applications, for example, access to EDI, X.400, X.25, ISDN, facsimile/PSTN, Internet (perhaps with recommended minimum rates so that the content providers can guarantee some quality of service for their applications),

- execution requirements for loading content — parameters will include any unpacking of files, any mapping of logicals to physicals, whether provided content includes, for example, any derived information such as visible FF or FR information to show fast forward/fast reverse of the linear audio-visual information,

- on content coming from a telecommunications link, objects are needed at some stage to define whether the material is to be made available immediately (broadcast), and if so, whether the material is also able to be stored for later retrieval by customers (purely transitory or not, and if not, for how long it can be retained), the encoding and multiplexing employed on the link, and whether the material is arriving in real time — on live links, the objects describing the material may well arrive on a separate physical or logical link, but that makes little difference to the definitions (there may well be specially defined 'live broadcast' parameters which are used by default, since a live broadcast, by definition, cannot be stopped and restarted from the top),

- definition of class of material, in terms of predefined groups able to use it — parameters might include (for entertainment) class of viewing certificate, payment band, availability by time of day and/or after a given release date and/or rendered unavailable after a specific date/time; for other material, parameters might include predefined CUG classes (e.g. schools, educational establishments, registered hospitals only),

- descriptive and menu information for the content, such that any picture icons, descriptive text, major names, subject matter, etc, can be automatically entered into search databases, and content navigation systems and menus on the server.

4.2 Small-scale interactive broadband servers

Smaller-scale broadband servers tend to be directed more towards the films/TV programmes-on-demand services, and so they can be focused much more closely on the small number of users or sessions, and the specific application for which they will be used. Applications are already appearing not only in trial cable-TV systems, but also in hotel systems and airline systems. The controlling software tends to be a lot simpler, so the audio-visual pump and controller tend to be integrated, and are based on a PC of some form and/or a workstation. Some systems are emerging from disk integrators, exploiting their own knowledge of the detailed performance parameters of magnetic disk systems.

4.3 Staggercast broadband servers

A staggercast server (see Fig 6) works on the basis of broadcasting a small number of popular titles, with each title broadcast a number of times, each time staggered by, say, 5—10 minutes, so that a user, on requesting a title, simply has to wait for the next start time, up to the maximum stagger time used by the system. In such a system, users can 'pause' their viewing, and then take up viewing on the next delayed version of the same title which most closely matches their pause position, and can 'fast forward' and 'fast reverse' by browsing through the other broadcasts of the same title, but only in multiples of the stagger time. The service to users can only consist of a few popular titles, and, in general, real transactional services are not available.

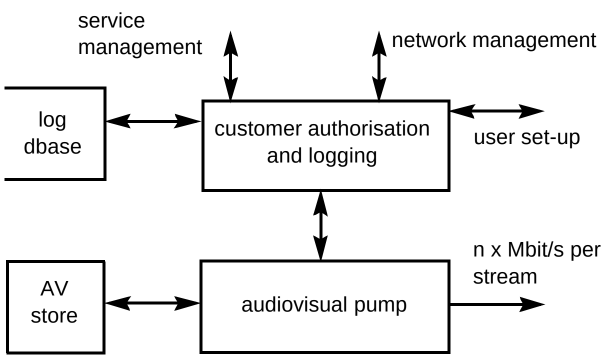

Fig 6 A typical staggercast broadband server.

If a server is assumed to be offering 100 channels carrying 20 films, for a 2-Mbit/s system this would imply:

- approximately 200 Mbit/s downstream capability,
- 30 Gbyte on-line store.

This is clearly a much simpler task than the fully interactive broadband server, and the server needs to have little interactive capability, save managing user log-ins and billing, and the crude navigation outlined above.

Implementation of such a server can take various routes, including simply cutting down an interactive broadband server defined above. One alternative is to exploit the prior scheduling of the spooling activity and use storage media that are cost-effective for this, like tape. One type of tape system which can be of interest is that used for digital studios, namely D1 or D3. As these tapes provide user information at 270 Mbit/s, one machine could be used to spool out 130 channels (at 2 Mbit/s) in parallel, for the length of the tape, which is usually 45 minutes at maximum. Two machines can be readily synchronised so the second takes over from the first to provide a seamless join. In addition, as small- and large-scale mechanical tape-handling systems already exist for the broadcast station market, the loading can be entirely automated. Special systems will be required to master the original tapes (including all the staggering of the showings of the films), but this is not a major issue. Churning of the information will require new tapes being produced with the new correct mix of information. Such tape systems would only be loosely coupled to the server control to report status and control at a high level, programme tape loading, and timing. Broadcast tape machines are relatively expensive, but are designed for high availability and low maintenance cost, and few are needed for such a system.

5. Low bit rate servers

For completeness, a brief discussion on low bit rate (LBR) servers is included. As defined earlier, LBR servers aim to address services requiring $n \times 64$ Kbit/s user streams, where n = 1—6. Where n = 1—2, this can most easily be provided over standard ISDN and so can be used directly on BT's existing network; additional access network considerations may be required where n approaches 6 (384 Kbit/s). A typical LBR server is depicted in Fig 7.

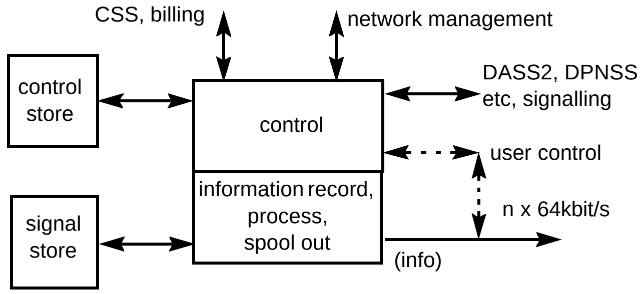

Fig 7 An LBR server.

The speech applications platform (SAP) already used in the BT network, is such a server, initially providing speech-based services (Callminder, etc) [4]. Most such servers are based on a flexible architecture, so additional services can be added later by software changes and without changing the installed hardware system.

These types of platform are already being extended mostly by additional software, to provide other LBR services such as telemetry, facsimile, new chargecard-type services, as well as enhanced speech services. With some additional processing capability and software, they could also be enhanced to provide the audio-visual and multimedia equivalent of many of the above, including videophone database, videophone 'Callminder', VC8000 database dial-in, and dial-in audio-visual information services where these do not contravene the regulations regarding broadcast services.

As already mentioned in earlier sections, limited audio-visual services could be provided from such systems. At lower (ISDN2) bit rates, small-size moving images, full-screen graphics and still images can be used to good effect for a number of interactive multimedia services. Current telephony tariffing might restrict the use of such a system for long hold-time sessions from residential users, as it would appear that they would have to make at least a local call equivalent to one or two PSTN phone calls to access the server. The discussion on offering differential network tariffs for different services is not within the scope of this paper.

Instead of providing audio-visual material, an alternative scenario could be to use such network-based LBR servers to provide authorisation, perhaps some limited interactivity, and logging of events for services whose audio-visual element is provided over another means, such as digital satellite, cable-TV or digital terrestrial TV. In this situation the LBR server effectively replaces the server control part of a typical staggercast broadband server (see section 4), but uses the existing BT network for control and interactivity communications from (and possibly to) the customer.

6. Conclusions

There is unlikely to be one technical solution for all VoD server requirements, as these can vary considerably, dependent on overall service requirements.

Large-scale interactive broadband servers still present a considerable technical challenge to server suppliers, but technical solutions are emerging, and cost trends would indicate that commercially viable solutions will be possible in the near future.

Servers with less functionality than the fully interactive broadband server can be provided at considerably less cost per user stream, but the reduction in functionality may not be acceptable for all planned interactive TV services.

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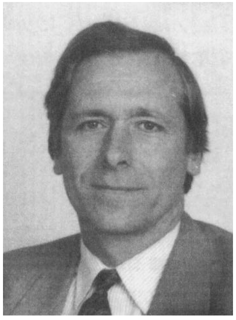

Gordon Kerr joined BT Laboratories in 1975 after graduating from Cambridge University in Electrical Sciences. For six years he worked on BT’s first digitally implemented PSTN modem, and then founded a new group to define, design, develop and install a video library service on the Westminster Cable-TV system. In 1987 he took on a systems section in the video telecommunications area, which eventually included laying the major technical foundations for BT’s current interest in video on demand, with more recent emphases on the video server and large-scale audiovisual encoding for the BT interactive TV trials. In 1995 he moved from the delivery part of BT Laboratories into a more research-oriented area to pull together and lead the medium and longer-term work within BT Laboratories on interactive multimedia services.

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